

Portfolio Optimization Primer

Energy Sector | 2/22/2026

Overview of Max Diversification and Hierarchical Risk-Parity

Maximum Diversification Model (MDP)

Overview of Optimization Techniques

The Maximum Diversification Portfolio (MDP) constructs an optimal portfolio in the energy sector by maximizing diversification, using asset volatilities and correlations rather than expected returns. This is well suited to energy stocks, which are often highly correlated due to shared exposure to commodity prices and macro factors. By emphasizing low-correlation assets, the model aims to reduce concentration risk and improve stability. However, it relies on accurate covariance estimates and can be sensitive to changing market conditions. In practice, this results in meaningful reallocations toward assets that contribute the most to diversification while reducing exposure to more correlated or redundant holdings, often leading to a more uneven but risk-efficient distribution of weights.

Economic Interpretation

Because of its structural design, the MDP portfolio reflects diversification benefits driven by volatility and correlation rather than factor exposures. In this energy-sector portfolio, the optimizer increases exposure to NEE (+19.34%), CEG (+2.84%), and FRO (+2.29%), while significantly reducing positions in WDS (-10.99%), FANG (-8.31%), and AROC (-5.17%). These shifts are driven purely by the covariance structure of the assets, favoring securities that improve overall diversification and reduce portfolio risk. While this approach enhances risk distribution, it can still lead to concentration, most notably in NEE, and remains sensitive to changes in correlations within the sector.

Hierarchical Risk-Parity (HRP)

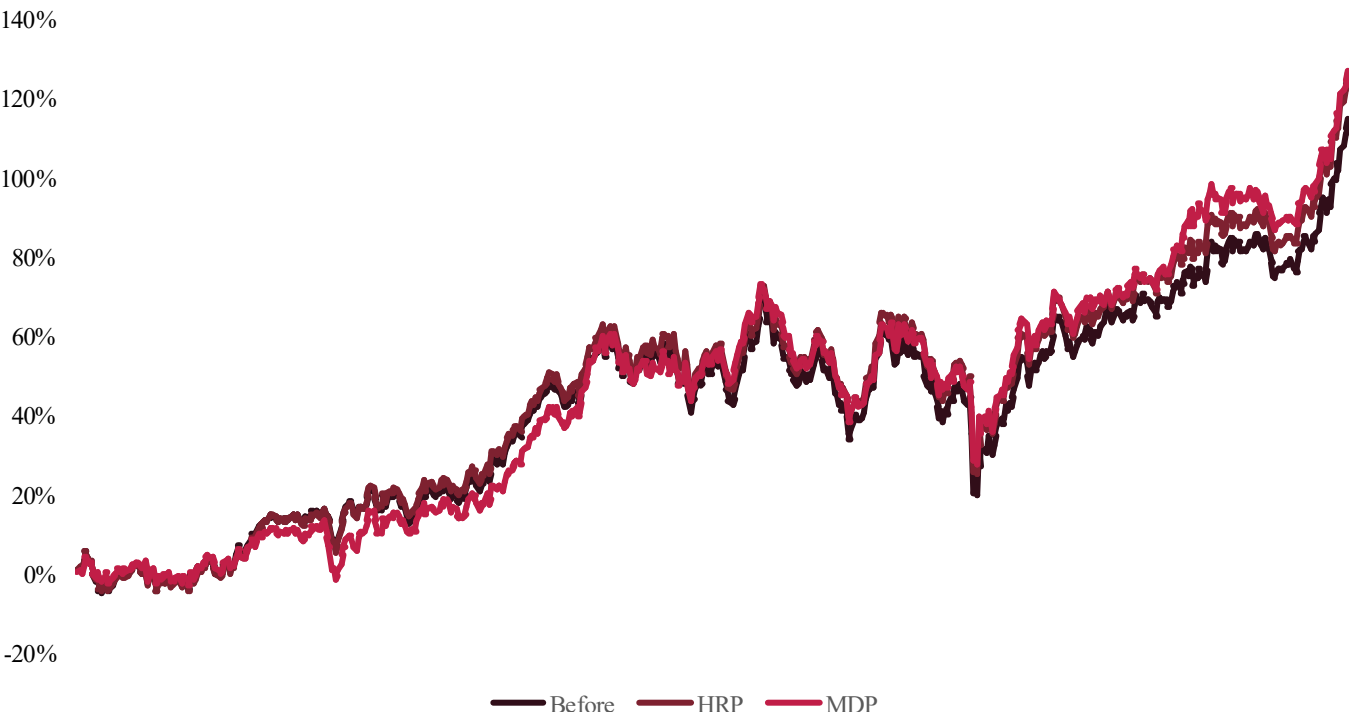
Overview of Optimization Techniques

In the energy sector, where equities often move together due to shared exposure to commodity prices, geopolitical dynamics, and macroeconomic conditions, HRP helps limit concentration risk by allocating capital across clusters of related firms rather than individual securities. By grouping energy companies, such as upstream producers, midstream operators, and integrated firms based on their historical correlation structure, HRP improves diversification relative to traditional Risk Parity while reducing reliance on precise pairwise correlation estimates. However, portfolio outcomes still depend on the chosen clustering methodology, and different grouping choices can lead to meaningfully different allocations.

Economic Interpretation

By clustering stocks based on their co-movement, HRP limits risk concentration across highly correlated energy subsectors, leading to more balanced exposure across utilities, upstream producers, and energy services. In this portfolio, HRP increases allocations to NEE (+2.06%), FRO (+2.84%), and CEG (+0.52%). At the same time, it reduces weights in WDS (-3.39%), FANG (-1.44%), and AROC (-0.59%), reflecting their stronger correlation within commodity-sensitive clusters. These adjustments are driven by correlation structure rather than fundamental views, helping reduce concentration in closely related energy names. By reallocating based on evolving correlation patterns instead of fixed volatility assumptions, HRP promotes a more resilient and balanced exposure across changing market conditions.

Cumulative Returns Over Time (3-Year Return)



Portfolio Optimization Primer

Energy Sector | 2/22/2026

Overview of Mean Variance and Black-Litterman

Mean Variance

Overview of Optimization Techniques

Mean-Variance optimization chooses portfolio weights by balancing two goals: higher returns and smaller ups and downs. Rather than weighing everything equally, it accounts for how each stock behaves and how the stocks move together, then favors combinations that have offered a better tradeoff between reward and stability. Its strengths are that it follows a clear, widely used framework, encourages diversification by considering relationships between holdings, and provides a structured way to be more aggressive or conservative. Its weaknesses are that small input changes can lead to different results, it can concentrate too much on a few names without limits, and it can struggle when market conditions shift.

Economic Interpretation

Because of its heavy reliance on historical return data, Mean-Variance Optimization exhibits a strong bias toward assets with recent momentum while aggressively penalizing those with lagging historical metrics. In this energy portfolio, the optimizer completely abandons international commodity exposure, cutting allocations to Woodside Energy (-20.17%) entirely to zero. Instead, it equally concentrates capital into the remaining domestic utilities, drillers, and midstream assets, pushing Constellation Energy (+7.84%), Diamondback Energy (+5.28%), and Archrock (+2.16%) perfectly to the 20% position limit. These shifts are driven by the backward-looking assumption that past price inefficiencies and localized outperformance will persist indefinitely. Mean-Variance works best when historical correlations remain stable, but it fails here by completely eliminating international diversification (WDS) simply because of a poor historical return window, ignoring the global nature of energy markets.

Black-Litterman

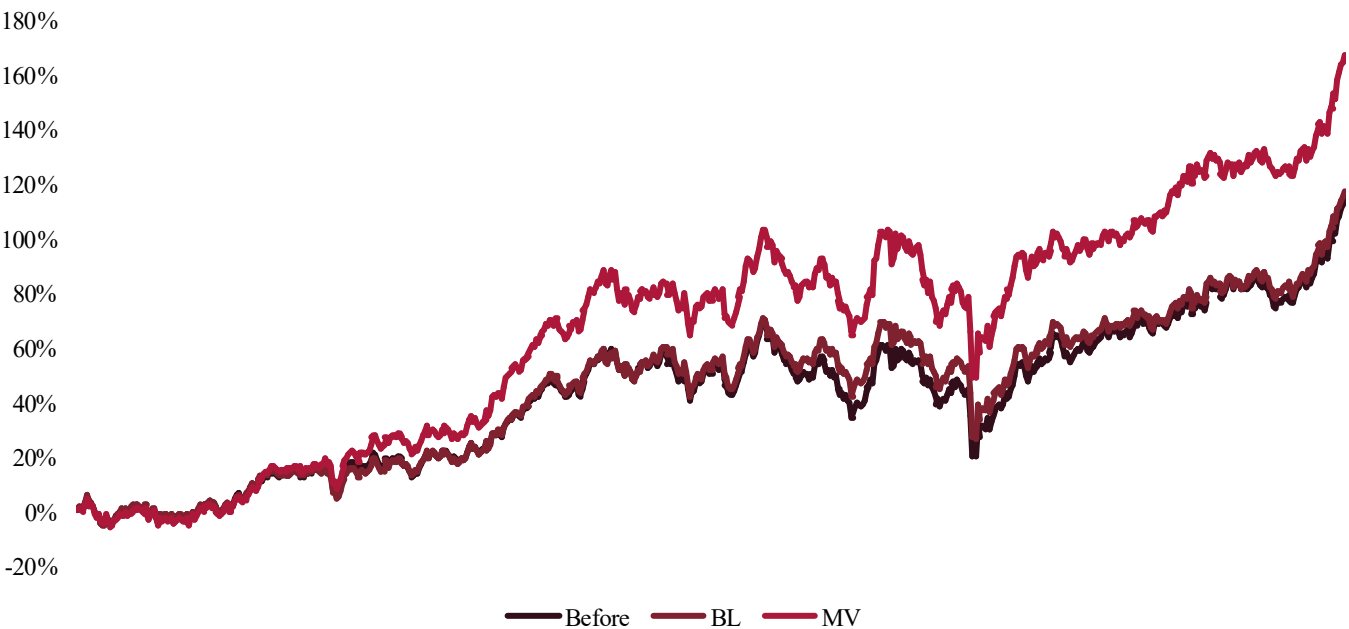
Overview of Optimization Techniques

Sentiment-Enhanced Black-Litterman builds on the standard Black-Litterman approach by combining two sources of “what we think will happen next”: the team’s own views and the tone of recent financial news. In practice, it reads current headlines about each company, summarizes whether the coverage is mostly positive or negative, and then uses that information to gently tilt the portfolio toward stocks with stronger sentiment and away from those with weaker sentiment, while still keeping the overall portfolio diversified. The benefit is that it makes the process less dependent on one person’s opinion and more responsive to new information that may be emerging in the market. The tradeoff is that it only works well if the news is reliable and consistently available, and it adds extra moving parts that can amplify noise when headlines are misleading, overly hyped, or intentionally manipulated.

Economic Interpretation

Because of its ability to integrate forward-looking sentiment and thematic views, Black-Litterman exhibits sensitivities to market narratives rather than just past price action. In this energy portfolio, the Black-Litterman optimizer heavily favors companies with strong fundamental catalysts, maximizing allocations to Constellation Energy (+7.84%) and Archrock (+2.16%) to the 20% limit. Conversely, it actively reduces exposure to traditional renewable utilities and domestic drillers, cutting NextEra Energy (-6.94%) and Diamondback Energy (-1.82%), while notably saving Woodside Energy (17.99%) from the total exclusion seen in the standard model. These shifts are driven by the specific confidence levels generated by sentiment models, which prioritize forward-looking catalysts, such as nuclear AI data center demand for Constellation, over purely historical variance. Black-Litterman works best here by pricing in current market enthusiasm and news flow, allowing the portfolio to proactively position for secular themes rather than reactively maximizing historical math.

Cumulative Returns Over Time (3-Year Return)

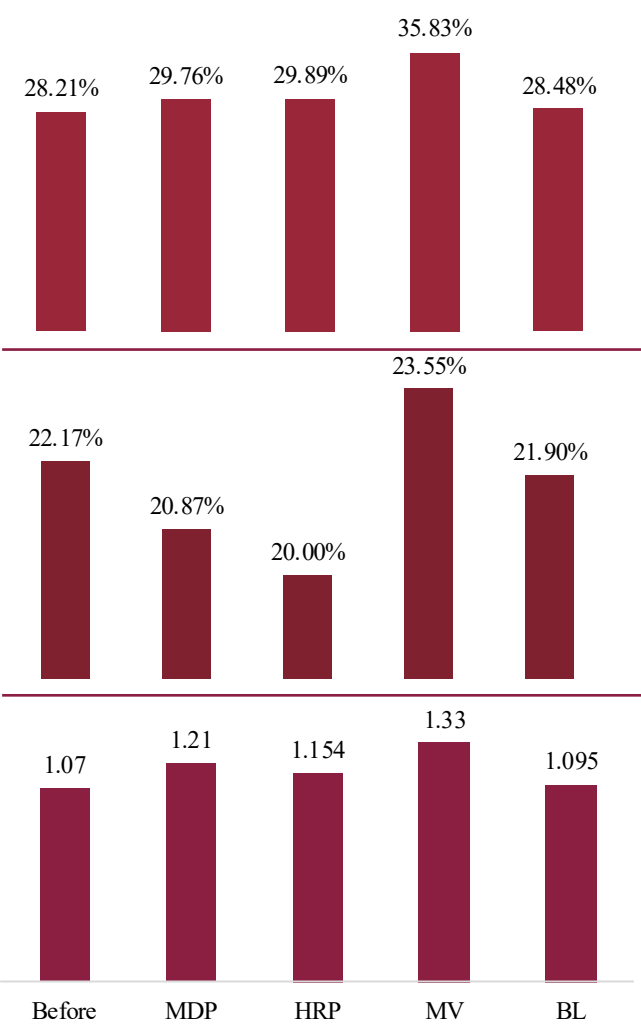


Portfolio Optimization Primer

Energy Sector | 2/22/2026

Comparative Analysis

Optimizer	Annual Return	Annual Vol	Sharpe Ratio	Sortino Ratio
Before	28.21%	22.17%	1.070	1.540
MDP	29.76%	20.87%	1.210	1.765
HRP	29.89%	20.00%	1.154	1.664
MV	35.83%	23.55%	1.330	1.951
BL	28.48%	21.90%	1.095	1.556



Expected return (Annual Return)

Highest (MV): 35.83%

Lowest (BL): 28.48%

Interpretation: The Mean-Variance model achieved the highest return (35.83%) by aggressively concentrating capital into domestic assets with strong recent momentum, eliminating lagging international plays. In contrast, the Black-Litterman model produced the lowest return (28.48%) because its forward-looking sentiment views tempered this historical bias, sacrificing backtested yield for balanced global exposure.

Volatility (Annual Vol)

Lowest (HRP): 20.00 %

Highest (MV): 23.55%

Interpretation: Hierarchical Risk Parity (HRP) achieved the lowest volatility (20.00%) by correctly grouping distinct business models, separating stable utilities from volatile shipping to balance structural risks. The Mean-Variance model exhibited the highest volatility (23.55%) because it ignored these differences, concentrating in highly correlated, high-beta assets to maximize historical growth.

Sharpe ratio (Risk Adj)

Highest (MV): 1.154

Lowest (BL): 1.095

Interpretation: The Mean-Variance model posted the highest Sharpe Ratio (1.330) by overfitting to the recent outperformance of domestic energy names, assuming those trends will repeat. The Black-Litterman model produced the lowest ratio (1.095) because it diverged from this backward-looking efficiency, accepting a lower historical stat to position for future sentiment-driven catalysts.

Analysis:

After analyzing the optimization models, NEE received the strongest allocation, reaching 37.28% in MDP and the 20% cap in HRP, highlighting its role as a key diversifier. In contrast, FANG had one of the lowest average allocations (13.15%), with a sharp reduction in MDP (6.41%) due to its higher correlation with commodity-sensitive assets. FRO showed the highest average allocation (19.39%), reflecting consistent importance across models, while CEG, AROC, and WDS remained more stable (~16.92%), indicating moderate contributions to diversification without dominating portfolio construction. These differences highlight how each model reallocates capital based on structure, favoring diversification and correlation benefits over concentrated exposures.

Capital Allocation For Each Stock					
Tickers	Before	HRP	MDP	MV	BL
CEG	12.16%	12.68%	15.00%	20%	20.00%
FANG	14.72%	13.28%	6.41%	20%	12.90%
FRO	17.16%	20.00%	19.45%	20%	18.11%
NEE	17.94%	20.00%	37.28%	20%	11.00%
AROC	17.84%	17.26%	12.67%	20%	20.00%
WDS	20.17%	16.78%	9.18%	0%	17.99%

Portfolio Optimization Primer

Energy Sector | 2/22/2026

Recommended Optimization Framework: Hierarchical Risk-Parity (HRP)

Hierarchical Risk Parity (HRP) is applied to the energy sector portfolio to construct a balanced and resilient weighting scheme based on the correlation structure of assets. This approach avoids reliance on precise covariance estimates or return forecasts, reducing sensitivity to estimation error and extreme assumptions that can distort traditional optimizers. By clustering energy stocks according to their co-movement, HRP allocates capital across distinct groups, helping to limit concentration in highly correlated, commodity-sensitive names. While HRP may produce slightly lower returns than more aggressive optimization methods, it delivers more stable capital allocations and relatively fair annual volatility. This results in a portfolio that is less reactive to short-term market noise and more robust across changing energy prices and macroeconomic conditions.

Fit with the Nature of Energy Businesses

This framework fits energy businesses particularly well because the sector is heavily influenced by shared macro drivers, including commodity price movements (FANG, WDS), demand and transportation dynamics (FRO), capital intensity and services exposure (AROC), and regulated or renewable stability (NEE, CEG). This is reflected in the allocation, where HRP maintains meaningful exposure to diversifying assets like NEE (20.00%) and FRO (20.00%), while keeping balanced positions across more commodity-sensitive firms such as CEG (12.68%), AROC (17.26%), and WDS (16.78%). FANG (13.28%) is more moderately weighted, reflecting its stronger correlation within upstream clusters. By allocating based on correlation structure rather than return expectations, the portfolio better captures how energy stocks respond differently to shifts in oil prices, demand cycles, and macro conditions, while avoiding unintended concentration in a single risk factor.

Fit with the Macro Environment

The energy landscape is currently defined by divergent drivers, including global commodity cycles and localized utility demand from artificial intelligence. Standard investment formulas struggle here because they treat all volatility as identical risk, often abandoning volatile international producers or blindly chasing domestic momentum. Our Hierarchical Risk Parity approach fixes this by mathematically grouping assets into distinct business clusters, separating stable utilities from erratic shipping and drilling. This builds a portfolio resilient to specific sub-sector shocks rather than one heavily concentrated in a single macro trend. The result is a portfolio that achieved a nearly 30% return by participating in utility growth while maintaining structural balance through global diversification.

Practical Implications for Portfolio Construction

In practical terms, this framework improves position sizing by determining weights through a cluster-based risk lens before deploying capital. For example, where a Mean-Variance model eliminated international exposure by cutting WDS to 0% to chase historical efficiency, the HRP model settles on a balanced 16.78% position. From a risk management standpoint, the framework prevents any single economic driver from dominating the portfolio, resulting in high-conviction allocations to defensive utilities like NEE (20.00%) and uncorrelated shipping assets like FRO (20.00%) while keeping a modest allocation to domestic drillers like FANG (13.28%). This clustering approach grounds allocations in fundamental business differences rather than pure price variance, resulting in a Sharpe Ratio of 1.154 that successfully balances historical growth with true diversification.

Limitations and Caveats

However, there are important limitations to consider. The Hierarchical Risk Parity model relies on historical correlations to build asset clusters, meaning sudden macro shocks that correlate shipping and utilities could render its diversification illusory. As a purely backward-looking model, it cannot incorporate high-conviction, forward-looking catalysts like specific AI power contracts or geopolitical supply disruptions. Finally, prioritizing risk balance often sacrifices maximum returns and assumes historical asset structures remain stable, an assumption that can easily break down during a global liquidity crisis or sudden commodity collapse.

Cumulative Returns Over Time (3-Year Return)

